

Linear Temporal Logic

Andrew Nguyen



What is Temporal Logic?

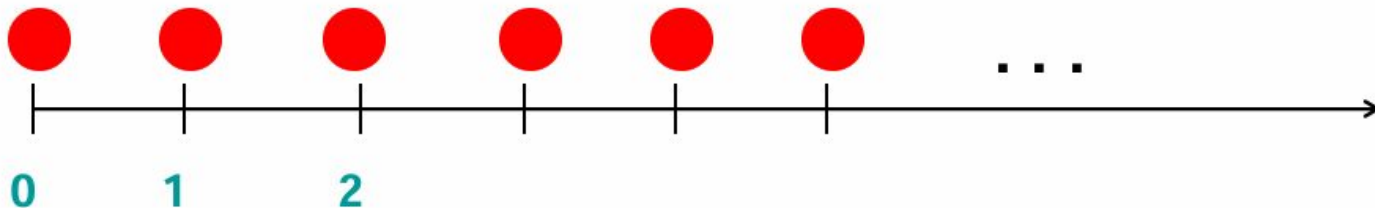
- Logic for specifying properties over time
 - Ex. Behavior of finite-state system
- One case: Propositional temporal logic
 - Other logics include:
 - Real-time temporal logic
 - Metric temporal logic
 - Signal temporal logic

Linear Temporal Logic Properties

Globally (Always) p : $G\ p$

- Means $G\ p$ is always true for a computation path if p always holds along the path

Suppose $G\ p$ holds along the path below starting at s_0

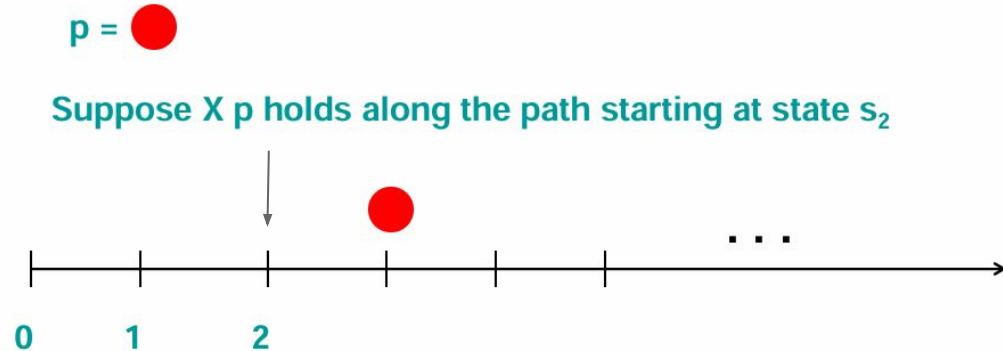


Eventually p: $F p$

- $F p$ is true for a path if p holds at some state along that path
- Means p just has to be true once or more on a path for $F p$
- Ex. $F (\text{Sun Sets}) \wedge ((\text{Sun Sets}) \rightarrow G (\neg(\text{Sun Sets}) \vee (\text{New Day})))$
 - The sun sets at some point during the day
 - Once the sun sets, it's the end of the day or it is a new day

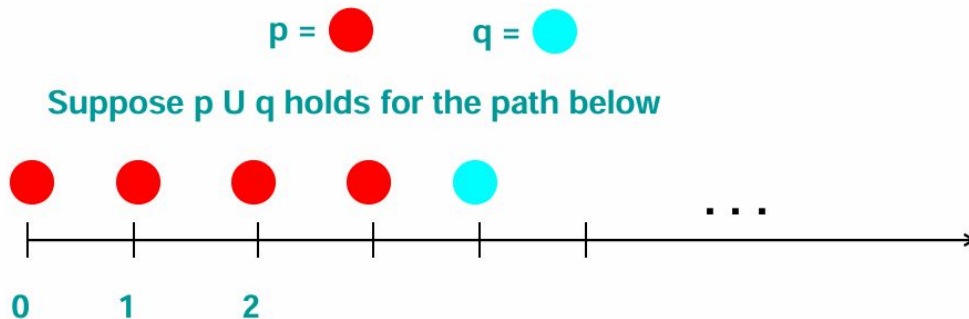
Next p: X p

- X p is true on paths starting in state s_i if p holds in next state s_{i+1}
- Essentially means that p will be true in the state right after the current state



$p \text{ Until } q: p \text{ U } q$

- You might notice $p \text{ U } q$ resembles $p \text{ Union } q$
- In Linear Temporal Logic, it is slightly different
- $p \text{ U } q$ is true along path starting at s if
 - q is true in some states reachable from s
 - p is true in all states from s until q holds



Formula Nesting

- p itself can also be a temporal logic formula
- Allows nested logic statements